Sahaj Bhadja

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Summary

Software Engineer with 2+ years building production AI services, fault-tolerant orchestration, distributed APIs, and containerized infrastructure using Python, C++, PyTorch, FastAPI, Kubernetes, Docker, AWS, Azure, CI/CD, automated testing, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Distributed Systems, Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Restructured fault-tolerant asynchronous LLM orchestration with explicit job states, retries, and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading service failures.
- Architected multi-tenant data infrastructure on Azure SQL and Blob Storage for 500+ users, reducing indexed-query latency from 1.2 seconds to 350 milliseconds through schema, indexing, and access-pattern optimization.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent AI-processing workflows with multithreading and secure ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, and recoverable execution under load.
- Implemented production inference-backed services handling 10K+ daily requests for 150+ users, applying automated tests and CI/CD deployment workflows while consistently maintaining stable API performance during peak usage.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a PyTorch simulation and evaluation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible data transformations, benchmark automation, regression testing, and documented experiment results.
- Developed a spatial-temporal encoder combining map geometry and agent context, improving closed-loop stability 15% across complex scenarios through systematic failure analysis, controlled evaluation, and root-cause debugging in Python.

Aasma Technology Solutions **August 2023 – July 2024**
Software Development Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments consistently.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates consistently across production workflows.
- Automated deployments with Jenkins CI/CD and Docker packaging, achieving 95%+ successful build validation and cutting release time 50% through repeatable configurations, staged testing, and documented deployment processes.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash, YAML
ML & Services: PyTorch, Transformers, LLM Systems, FastAPI, Flask, REST APIs, Microservices, Async Workflows, OOP
Infrastructure: Kubernetes, Docker, AWS, Azure, GCP, Linux, CI/CD, Jenkins, GitHub Actions, Cloud Storage, PostgreSQL
Reliability & Performance: Distributed Systems, State Transitions, Retries, Failure Isolation, Unit Testing, Integration Testing, Debugging, Monitoring, Performance Optimization

Projects

CaptionCraft - Model Inference Platform Live | *PyTorch, Transformers, Flask*

- Benchmarked BLIP, GIT, and ViT-GPT-2 across 10K images and 100K captions, improving measured accuracy from 81% to 96% while building scalable Flask inference APIs for real-time generation.

CPU Scheduling Simulator Live | *C++, Scheduling Algorithms, React*

- Implemented FCFS, SJF, SRT, round-robin, priority, and multilevel-queue scheduling with dynamic turnaround and waiting-time calculations across 1K+ process combinations, applying performance engineering to resource-allocation tradeoffs under load.

FinNexus Live | *Kubernetes, TypeScript, MongoDB, CI/CD*

- Built a Kubernetes-backed platform supporting 50K+ users and 1M+ transactions with secure integrations, real-time synchronization, automated deployment workflows, advanced filtering, and reliable containerized production infrastructure at scale.